COURSE NAME : PLASTIC MOULDING TECHNOLOGY
COURSE CODE : 5113
COURSE CATEGORY : E
SEMESTER : 5
PERIODS / WEEK : 4
PERIODS/SEMESTER : 60
CREDITS : 4

TIME SCHEDULE

MODULE	TOPICS	PERIODS
1	Plastic materials and processing techinques	15
2	Elementry mould design and injection moulding	15
3	Intermediate mould design	15
4	Compression,Blow and Transfer mould design Plastic product design and Decoration of plastic products.	15
	Total	**60**

COURSE OUT COMES:

On completion of the course the student should be able to:-
1.Understand different plastic materials ,properties and plastic processing methods.
2.Know the design procedure and working of Injection moulding machine.
3.Comprehend intermediate Injection mould design.
4.Understand compression,Blow and Transfer mould design.

SPECIFIC OUTCOMES

MODULE: 1

1.1.0 Understand different plastics materials, properties & plastic processing methods

1.1.1 Identify different plastics materials and their properties
1.1.2 Illustrate the conventional injection-moulding machine types, their specification, operation terminology and their parts
1.1.3 Compare different moulding processes used in industries, their application

MODULE: 2

2.1.0 Know the design procedure and working of injection moulding machine

2.1.1 List the design procedure for injection moulding
2.1.2 Explain the working of injection moulding machine

MODULE: 3

3.1.0 Comprehend intermediate injection mould design

3.1.1 Identify the elements of intermediate injection moulds
3.1.2 Explain the different types of intermediate injection moulding machine

MODULE: 4

4.1.0 Understand compression, blow and transfer mould design

4.1.1 Explain the design procedure for compression moulding
4.1.2 Justify the concepts in the design of blow moulds
4.1.3 Illustrate the decoration techniques, plating techniques used for plastic components

CONTENT DETAILS

MODULE:1 PLASTIC MATERIALS AND PROCESSING TECHNIQUES

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Plastic processing methods (concepts & description only)
Injection moulding: hot runner injection moulding process, gas assisted injection moulding process , multi color and multi component injection moulding process , reaction injection moulding. Compression moulding – bulk factor, compression moulding procedure, types of compression mould (Flash, fully positive, semi positive, and landed positive mould), difference between injection and compression moulding. Transfer moulding – pot transfer and plunger transfer moulding, advantages and disadvantages.
Blow moulding: extrusion blow moulding, injection blow moulding and stretch blow moulding, rotational moulding, thermoforming, pultrusion. Extrusion – pipe extrusion, blown film, cast film extrusion, rod extrusion – co extrusion , lamination techniques – types(wet, dry, thermal or heat, wax or hot melt and extrusion lamination)

MODULE:2 ELEMENTARY MOULD DESIGN AND INJECTION MOULDING
Injection Moulding Machines : Basic part and functioning of an injection moulding machine , types of injection moulding machine(screw type & plunger type) – single stage and two stage – clamping unit(toggle & hydraulic)- types of nozzles – typical injection moulding cycle, cycle time – machine specifications(definition only)
Elementary mould design: Basic terminology of injection moulds – bolsters, requirements , types – cavity and core inserts - selection of guide pillars and bushes – design of sprue bushes and registering– shrinkage calculation – core and cavity dimension.

Feed system: runner, cross section shape, runner size , runner layout – gates necessity, center gate, edge gate , balanced gating, types of gates - ejection system: ejector grid – ejector plates assembly – ejector rod, ejector plate and ejector retaining plate - methods of ejection – ejection from fixed half sprue puller

Parting Surface : Flat parting surface – non flat parting surface – venting – mould clamping – direct, indirect – cooling system - cooling integer type cavity plates – cooling integer type core plate – cooling bolster – cooling cavity inserts – cooling core inserts – water connection and seals (Concept & description only)

MODULE:3 INTERMEDIATE INJECTION MOULDING DESIGN

Moulding external undercuts: Split Mould – finger cam, dog leg cam & track. Hydraulic & spring actuation of split – side core and side cavity, method of actuation, Mould with internal under cut. Form pin , actuation , split core , jumping off – Mould for threaded component. Manual & automatic unscrewing methods hand mould for rotating lose core methods – multi daylight mould – under feed mould – Triple day light mould – hot runner unit mould, advantages and limitations , hot runner, manifold & its type. Hot runner nozzles & sprue, runner less mould – materials for injection mould – standard mould systems, advantages and limitation , popular manufactures of mould units (Concept & description only)

MODULE:4 COMPRESSION , BLOW AND TRANSFER MOULD DESIGN

Compression mould design : Economic determination of number of cavities , flash thickness allowances , design of mould cavity ,design of loading chamber , bulk factor , loading chamber depth &heat requirement for heating the mould related to - curing time , breathing time, Materials for compression mould.

Blow mould design : Materials selection , mould cooling, clamping force, mould venting, pinch-off, head die design, Parison diameter calculation , wall thickness, vertical – load strength, blow ratio, base pushup, highlights, rigidizing, shapes, design based considerations – shrinkage , neck and shoulder design, thread and beads, bottom design. Materials for blow mould.

Transfer mould design : Design of pot and plunger , runner & gate dimensions, bulk factor Feed system, Economic determination of the number of cavities, design of mould cavity, design of loading chamber, heat losses and energy requirement to heat the mould. Materials for transfer mould. (concept & Description of design only)

Plastic product design: Wall thickness – ribs and profiled structures – gussets or support ribs – bosses – holes – radii & corners – tolerances – coring – undercuts – draft angle

Decoration of plastic parts : painting and coating (Dipping, spraying and depositing) – metalizing (vacuum metallization, vacuum evaporation, sputtering) – plating– Flame and arc spraying – hot foil stamping – hot transfer - in mould decorating (Film insert moulding, in mould transfer decoration, powder coating) – Water transfer – printing (pad printing, screen printing, sublimation printing) - laser marking , vapour polishing.

TEXTBOOKS

1) Pye R.G.W “Injection Mould Design” Longman scientific & Technical 1989
2) Plastics Design and Processes by S.C. Sharma(Standard Publishers)
3) Injection Moulding 2nd Edition by Athulya. A.S(Multitech Publishing Co.
4) Fundamentals of Plastic mould design-Tata McGrawHill Edn Pvt.Ltd)

REFERENCE BOOKS

1) Injection Moulding Handbook – Dominick V Rosato and Donald V Rosato(CBS publishers and Distributers-Delhi)
2) Mills.NJ. Plastics. ELBS 1986.